

Taha Iqbal

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• [LinkedIn](#), [GitHub](#)

Bioinformatics software engineer with 4+ years building and operating production-grade NGS and genomic analysis pipelines at scale. Combines deep expertise in workflow engineering (Nextflow, Snakemake, Docker, AWS) with clinical bioinformatics experience in both regulated diagnostics and population-scale research settings. Proven track record delivering reliable, reproducible pipelines processing terabytes of sequencing data across 50,000+ whole genomes, and deploying clinical NGS systems against 40,000-genome reference databases. Pursuing MSE in Scientific Computing at UPenn (2026).

Core Competencies

- **Pipeline Engineering:** Nextflow, Snakemake; end-to-end NGS workflow development, optimization, and scaling for terabyte-scale datasets; reproducible, version-controlled pipeline architecture
- **Cloud & Infrastructure:** AWS (S3, EC2, Glacier), Docker, Singularity, SLURM/HPC; parallelized cloud and on-premise pipeline execution; CI/CD and version control with Git
- **NGS & Genomics:** DNA-Seq, RNA-Seq, WGS, GWAS; GATK, PLINK, BCF/VCF, Samtools, Bedtools, Bioconductor; variant QC, annotation, cohort harmonization, genome build liftover
- **Programming:** Python (pandas, NumPy, scikit-learn, Flask, FastAPI), R, SQL, Bash/Shell, C++, Java
- **Clinical & Regulatory:** CAP-compliant pipeline validation; SOP authorship; clinical sample processing; pathogen identification from NGS data

Professional Experience

- **Perelman School of Medicine, University of Pennsylvania** **Philadelphia, PA**
Bioinformatician II *2021–Present*
 - Built and maintained reproducible Snakemake and Nextflow pipelines for NGS and GWAS analysis across 50,000+ whole genomes, enabling scalable variant discovery, cohort-level association analyses, and downstream statistical modeling
 - Engineered parallelized, cloud-native bioinformatics workflows on AWS (EC2, S3, Glacier) and SLURM/HPC clusters, significantly improving runtime and scalability for terabyte-scale sequencing data processing
 - Developed modular pipeline components for variant extraction, annotation, filtering, genotype harmonization, and genome build liftover (hg18/hg19 to GRCh38), distributed via version-controlled, reproducible releases
 - Integrated agentic coding workflows using Claude Code to accelerate pipeline development, automate quality control steps, and improve code maintainability across the team
 - Collaborated with computational biologists, statisticians, and investigators to design analytical approaches, interpret results, and deliver publication-ready analyses for ongoing Alzheimer's disease genomics studies
 - Mentored junior researchers on pipeline development, reproducible research practices, and statistical workflows
- **Aperiomics** **Sterling, VA**
Clinical Bioinformatician *2020–2021*
 - Designed and deployed a production NGS pipeline for HPV strain identification, serving NIH and X4 Pharmaceuticals clients; processed FASTQ/BAM/CRAM data against a 40,000-genome reference database of bacterial, viral, and fungal genomes
 - Containerized all bioinformatics workflows using Docker and executed pipelines on AWS (S3, EC2, Glacier) and local HPC clusters, ensuring reproducibility, secure data access, and scalable clinical throughput
 - Automated data transfer, pipeline orchestration, and report generation workflows, measurably reducing turnaround time for clinical results delivered to physicians and patients
 - Conducted validation studies for fecal and plasma sequencing assays and authored SOPs and QC documentation in compliance with CAP clinical laboratory regulations

- **Temple University** **Philadelphia, PA**
Researcher *2018–2020*
 - Built and maintained a SQL database of multi-species protein and DNA sequences (FASTA) to support large-scale gene duplication and clustering analyses
 - Developed Python, R, and Bash scripts to automate retrieval and processing of biochemical data from public databases, improving reproducibility and reducing manual effort

Education

- **University of Pennsylvania** **Philadelphia, PA**
Master of Science in Engineering: Scientific Computing; Certificate in Biomedical Informatics *2024–2026*
GPA: 3.70/4.00
- **Temple University** **Philadelphia, PA**
Professional Science Master's: Bioinformatics *2019*
- **State University of New York at Albany** **Albany, NY**
Bachelor of Science: Biology *2016*

Selected Publications

- Rajabli, F., Benchek, P., Tosto, G. et al. (2025). Multi-ancestry genome-wide meta-analysis of 56,241 individuals identifies novel risk loci for Alzheimer's disease. *Genome Biol* 26, 210. [DOI](#)
- Kuzma, A., et al. (2024). NIAGADS: A Comprehensive National Data Repository for Alzheimer's Disease Genetics and Genomics Research. [DOI](#)

Honors and Awards

- **Second Place, 2025 Penn HealthCare Case Competition** – Blue Crane Bio Consulting (2025). Finalist team from 49 applicants across 44 global universities.

References

- Available upon request.